

HAOCHENG YIN

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RESEARCH OVERVIEW

My research goal is to develop intelligent embodied agents that are **generalizable** across diverse tasks and **adaptable** to unseen environments in the physical world. To address this goal, my research explores:

- Analyzing visual representations for robust robot control.
- Designing generalizable control modules for sim-to-real transfer.
- Building compositional generative models for effective world modeling.

Research Areas: Robotics, Machine Learning

EDUCATION

Georgia Institute of Technology Ph.D. in Robotics Advisors: Prof. Lu Gan and Prof. Yongsheng Chen	Atlanta, GA <i>August 2025 – Present</i> GPA: 4.00/4.00
ETH Zürich M.S. in Computer Science Major: Machine Intelligence	Zürich, Switzerland <i>September 2021 – October 2024</i> GPA: 5.27/6.00
University of Illinois Urbana-Champaign B.S. in Electrical Engineering ZJU-UIUC Dual Degree Program	Champaign, IL <i>September 2017 – May 2021</i> GPA: 3.94/4.00
Zhejiang University B.E. in Electrical Engineering & Automation ZJU-UIUC Dual Degree Program	Hangzhou, China <i>September 2017 – June 2021</i> GPA: 3.96/4.00

PUBLICATIONS

(* indicates equal contribution)

- [1] Han Qi, **Haocheng Yin**, Aris Zhu, Yilun Du, and Heng Yang. “Inference-Time Enhancement of Generative Robot Policies via Predictive World Modeling”. In: *IEEE Robotics and Automation Letters (RA-L)* (2026). DOI: [10.1109/LRA.2026.3673995](https://doi.org/10.1109/LRA.2026.3673995). arXiv: [2502.00622](https://arxiv.org/abs/2502.00622) [cs.R0].
- [2] Han Qi*, **Haocheng Yin***, and Heng Yang. “Control-oriented Clustering of Visual Latent Representation”. In: *International Conference on Learning Representations (ICLR)*. 2025. arXiv: [2410.05063](https://arxiv.org/abs/2410.05063) [cs.LG].

RESEARCH EXPERIENCE

Lunar Lab Advisors: Prof. Lu Gan and Prof. Yongsheng Chen Research Project: <i>Dynamics Learning on Deformable Objects</i>	Georgia Institute of Technology <i>August 2025 – Present</i>
Computational Robotics Lab Advisor: Prof. Heng Yang Master Thesis: <i>Understand and Improve Diffusion Policy for Robot Control</i> ICLR 2025 Spotlight: <i>Control-Oriented Clustering of Visual Latent Representation</i> RA-L 2026: <i>Inference-Time Enhancement of Generative Robot Policies via Predictive World Modeling</i>	Harvard University <i>February 2025</i>
Soft Robotics Lab Advisor: Prof. Robert Katzschmann Research Project: <i>Learning Behavior Priors for Dexterous Manipulation</i>	ETH Zürich <i>December 2023</i>
Optimization & Decision Intelligence Lab Advisor: Prof. Niao He Research Project: <i>Bio-plausible Meta Reinforcement Learning</i> Research Project: <i>Inverse Reinforcement Learning from Suboptimal Demonstrations</i>	ETH Zürich <i>September 2022</i>